

Multiple Choice (10 marks)

1. Which of the following has the greatest potential for compromising a user's personal privacy?
 - (a) A group of cookies stored by the user's Web browser
 - (b) The Internet Protocol (IP) address of the user's computer
 - (c) The user's e-mail address
 - (d) The user's public key used for encryption
2. Let $l = [1,2,3,4]$. What is $\max(l)$ in Python 3?
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
3. In Python 3, let $r = \text{lambda } q: q * 2$. What is $r(3)$?
 - (a) 2
 - (b) 6
 - (c) 3
 - (d) 1
4. In python 3, which of the following is floor division?
 - (a) /
 - (b) //
 - (c) %
 - (d) |
5. Which is the smallest asymptotically?
 - (a) $O(1)$
 - (b) $O(n)$
 - (c) $O(n^2)$
 - (d) $O(\log n)$

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the expression `['Hi!'] * 4` produces the list `['Hi!', 'Hi!', 'Hi!', 'Hi!']`.
7. True or False: In Python 3, accessing the first element of the tuple `('abcd', 786, 2.23, 'john', 70.2)` results in an error.
8. True or False: The boolean expression `a[i] == max || !(max != a[i])` simplifies to `a[i] != max`.
9. True or False: A large Java program that has been tested extensively may still contain bugs that were not detected during testing.
10. True or False: Python variable names are case-sensitive, meaning `'Variable'` and `'variable'` are considered different names.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the first duplicate element in a given array of integers.
12. Write a function to find three closest elements from three sorted arrays.

End of Paper